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Case driven design for implantable medical device

Abstract

A medical device includes: a case at least a portion of which functions as a first electrode; a second electrode disposed in a header coupled to the case; a core assembly, the core assembly including operational circuitry enclosed within a core assembly housing, wherein the case includes the core assembly housing; and a battery assembly, the battery assembly including a battery enclosed within a battery housing, where the case further comprises the battery housing; where the operational circuitry is configured to drive a regulated voltage onto the case.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION (1) This application is a continuation application that claims priority to U.S. patent application Ser. No. 15/930,755, filed May 13, 2020 and to U.S. provisional application No. 62/846,908, filed May 13, 2019, which are herein incorporated by reference in its entirety.

TECHNICAL FIELD

(1) Embodiments of the present disclosure relate to medical devices and systems for sensing physiological parameters. More specifically, embodiments of the disclosure relate to battery connection configurations of implantable medical devices.

BACKGROUND

(2) Implantable medical devices (IMDs) may be configured to sense physiological parameters and/or provide therapy and may include one or more electrodes for performing aspects of these functions (e.g., sensing electrodes). The overall usable volume enclosed within a housing of an IMD may be adjusted based on considerations of patient comfort and performance. Examples of IMDs include implantable cardiac monitors (ICMs), implantable loop recorders (ILRs), and the like, which can be configured to be subcutaneously implanted in a patient for monitoring one or more physiological parameters such as, e.g., physiological parameters associated with the heart and/or the lungs.

(3) To facilitate a more comfortable and efficient experience, these devices, which often are powered by an internal battery, may be designed to keep the overall volume of the device as small as possible. One way of reducing volume is to use the battery housing as part of the overall device case, as opposed to disposing the battery assembly (including the battery housing) within a device case. To further increase the efficient use of space in this type of device, one of the sensing electrodes may be electrically coupled to the battery housing and, in embodiments, the battery housing itself (or a portion thereof) may be used as the electrode.

SUMMARY

(4) In an Example 1, a medical device comprising: a case, wherein at least a portion of the case functions as a first electrode; a second electrode disposed in a header coupled to the case; a core assembly, the core assembly comprising operational circuitry enclosed within a core assembly housing, wherein the case comprises the core assembly housing; and a battery assembly, the battery assembly comprising a battery enclosed within a battery housing, wherein the case further comprises the battery housing; wherein the operational circuitry is configured to drive a regulated voltage onto the case. In an Example 2, the medical device of Example 1, the operational circuitry comprising a voltage regulator disposed between the battery and the case, the voltage regulator configured to drive the regulated voltage onto the case. In an Example 3, the medical device of Example 2, wherein the voltage regulator is configured to produce the regulated voltage by reducing a power supply voltage provided by the battery. In an Example 4, the medical device of either of Examples 1 or 2, further comprising a charge pump configured to facilitate increasing the regulated voltage to a voltage greater than a power supply voltage provided by the battery. In an Example 5, the medical device of any of Examples 2-4, wherein the first and second electrodes are coupled to an input component, and wherein the voltage regulator is configured to provide a power supply to the input component. In an Example 6, the medical device of any of Examples 2-4, wherein the first and second electrodes are coupled to an input component, and further comprising an additional voltage regulator, wherein the additional voltage regulator is configured to provide a power supply to the input component. In an Example 7, the medical device of any of Examples 1-6, wherein operational circuitry is configured to drive the regulated voltage onto the case in response to determining that the device is in an active mode. In an Example 8, the medical device of any of Examples 1-7, wherein the operational circuitry is configured to drive the regulated voltage at a level that is selected based on a battery discharge state. In an Example 9, the medical device of any of Examples 1-8, wherein the regulated voltage is positive. In an Example 10, a method of

manufacturing a medical device comprising: providing a case, wherein at least a portion of the case functions as a first electrode; couple a header to the case; dispose a second electrode in the header; provide a core assembly, the core assembly comprising operational circuitry enclosed within a core assembly housing, wherein the case comprises the core assembly housing; and couple a battery assembly to the core assembly, the battery assembly comprising a battery enclosed within a battery housing, wherein the case further comprises the battery housing; wherein the operational circuitry is configured to drive a regulated voltage onto the case. In an example 11, the method of Example 10, further comprising providing a voltage regulator disposed between the battery and the case, the voltage regulator configured to drive the regulated voltage onto the case. In an Example 12, the method of Example 11, wherein the voltage regulator is configured to produce the regulated voltage by reducing a power supply voltage provided by the battery. In an Example 13, the method of any of Examples 2-4, further comprising coupling the first and second electrodes to an input component, and wherein the voltage regulator is configured to provide a power supply to the input component. In an Example 14, the method of any of Examples 2-4, further comprising coupling the first and second electrodes to an input component, and further comprising providing an additional voltage regulator, wherein the additional voltage regulator is configured to provide a power supply to the input component. In an Example 15, the method of any of Examples 10-14, wherein operational circuitry is configured to drive the regulated voltage onto the case in response to determining that the device is in an active mode method of any of claims **10-14**, wherein operational circuitry is configured to drive the regulated voltage onto the case in response to determining that the device is in an active mode. In an Example 16, a medical device comprising: a case, wherein at least a portion of the case functions as a first electrode; a second electrode disposed in a header coupled to the case; a core assembly, the core assembly comprising operational circuitry enclosed within a core assembly housing, wherein the case comprises the core assembly housing; and a battery assembly, the battery assembly comprising a battery enclosed within a battery housing, wherein the case further comprises the battery housing; wherein the operational circuitry is configured to drive a regulated voltage onto the case. In an Example 17, medical device of Example 16, the operational circuitry comprising a voltage regulator disposed between the battery and the case, the voltage regulator configured to drive the regulated voltage onto the case. In an Example 18, medical device of Example 17, wherein the voltage regulator is configured to produce the regulated voltage by reducing a power supply voltage provided by the battery. In an Example 19, the medical device of Example 16, further comprising a charge pump configured to facilitate increasing the regulated voltage to a voltage greater than a power supply voltage provided by the battery. In an Example 20, the medical device of Example 17, wherein the first and second electrodes are coupled to an input component, and wherein the voltage regulator is configured to provide a power supply to the input component. In an Example 21, the medical device of Example 17, wherein the first and second electrodes are coupled to an input component, and further comprising an additional voltage regulator, wherein the additional voltage regulator is configured to provide a power supply to the input component. In an Example 22, the medical device of Example 16, wherein operational circuitry is configured to drive the regulated voltage onto the case in response to determining that the device is in an active mode. In an Example 23, the medical device of Example 17, wherein the operational circuitry is configured to drive the regulated voltage at a level that is selected based on a battery discharge state. In an Example 24, the medical device of Example 16, wherein the regulated voltage is positive. In an Example 25, a method of manufacturing a medical device comprising: providing a case, wherein at least a portion of the case functions as a first electrode; couple a header to the case; dispose a second electrode in the header; provide a core assembly, the core assembly comprising operational circuitry enclosed within a core assembly housing, wherein the case comprises the core assembly housing; and couple a battery assembly to the core assembly, the battery assembly comprising a battery enclosed within a battery housing, wherein the case further comprises the battery housing; wherein the operational

circuitry is configured to drive a regulated voltage onto the case. In an Example 26, the method of Example 25, further comprising providing a voltage regulator disposed between the battery and the case, the voltage regulator configured to drive the regulated voltage onto the case. In an Example 27, the method of Example 26, wherein the voltage regulator is configured to produce the regulated voltage by reducing a power supply voltage provided by the battery. In an Example 28, the method of Example 25, further comprising coupling the first and second electrodes to an input component, and wherein the voltage regulator is configured to provide a power supply to the input component. In an Example 29, the method of Example 26, further comprising coupling the first and second electrodes to an input component, and further comprising providing an additional voltage regulator, wherein the additional voltage regulator is configured to provide a power supply to the input component. In an Example 30, the method of Example 15, wherein operational circuitry is configured to drive the regulated voltage onto the case in response to determining that the device is in an active mode. In an Example 31, a medical device comprising: a case, wherein at least a portion of the case functions as a first electrode; a second electrode disposed in a header coupled to the case; a core assembly, the core assembly comprising operational circuitry enclosed within a core assembly housing, wherein the case comprises the core assembly housing; and a battery assembly, the battery assembly comprising a battery enclosed within a battery housing, wherein the case further comprises the battery housing; wherein the operational circuitry comprises a voltage regulator that is configured to drive the regulated voltage onto the case. In an Example 32, medical device of Example 31, wherein the voltage regulator is configured to produce the regulated voltage by reducing a power supply voltage provided by the battery. In an Example 33, the medical device of Example 31, further comprising a charge pump configured to facilitate increasing the regulated voltage to a voltage greater than a power supply voltage provided by the battery. In an Example 34, the medical device of Example 31, wherein the first and second electrodes are coupled to an input component, and wherein the voltage regulator is configured to provide a power supply to the input component. In an Example 35 the medical device of Example 31, wherein the first and second electrodes are coupled to an input component, and further comprising an additional voltage regulator, wherein the additional voltage regulator is configured to provide a power supply to the input component.

(5) While multiple embodiments are disclosed, still other embodiments of the present disclosure will become apparent to those skilled in the art from the following detailed description, which shows and describes illustrative embodiments of the disclosure. Accordingly, the drawings and detailed description are to be regarded as illustrative in nature and not restrictive.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a schematic illustration depicting a patient monitoring system, in accordance with embodiments of the disclosure.

(2) FIG. 2A is a perspective view of an implantable medical device (IMD), in accordance with embodiments of the disclosure.

(3) FIGS. 2B and 2C are partially-exploded perspective views of the IMD depicted in FIG. 2A, in accordance with embodiments of the disclosure.

(4) FIG. 3 depicts a conceptual circuit diagram depicting an illustrative circuit for an IMD, in accordance with the prior art.

(5) FIG. 4 depicts a conceptual circuit diagram depicting an illustrative circuit for an IMD, in accordance with the embodiments of the disclosure.

(6) FIG. 5 depicts a conceptual circuit diagram depicting an illustrative circuit for an IMD, in accordance with embodiments of the disclosure.

(7) While the disclosed subject matter is amenable to various modifications and alternative forms, specific embodiments have been shown by way of example in the drawings and are described in detail below. The intention, however, is not to limit the disclosure to the particular embodiments described. On the contrary, the disclosure is intended to cover all modifications, equivalents, and alternatives falling within the scope of the disclosure as defined by the appended claims.

(8) Although the term “block” may be used herein to connote different elements illustratively employed, the term should not be interpreted as implying any requirement of, or particular order among or between, various blocks disclosed herein. Similarly, although illustrative methods may be represented by one or more drawings (e.g., flow diagrams, communication flows, etc.), the drawings should not be interpreted as implying any requirement of, or particular order among or between, various steps disclosed herein. However, certain embodiments may require certain steps and/or certain orders between certain steps, as may be explicitly described herein and/or as may be understood from the nature of the steps themselves (e.g., the performance of some steps may depend on the outcome of a previous step). Additionally, a “set,” “subset,” or “group” of items (e.g., inputs, algorithms, data values, etc.) may include one or more items, and, similarly, a subset or subgroup of items may include one or more items. A “plurality” means more than one.

DETAILED DESCRIPTION

(9) Embodiments of the disclosure include an implantable medical device (IMD) in which a portion of the operational circuitry of the device is electrically coupled to the case of the IMD such that a sensing component senses signals between the case (housing) and an electrode disposed in the header of the device. In some previous devices, this was accomplished by using a case-negative design—that is, by shorting an anode terminal of the power circuitry to the case, which resulted in electrochemical issues such as, for example chemical reactions that caused cracking.

(10) To address the electrochemical issues, alternative solutions have included using a case-positive design. In the case-positive design, the case is shorted to the positive battery terminal. This case-positive design appeared to resolve the electrochemical issue but caused injection of a battery voltage ripple as a common mode noise that needed to be rejected by the sense filter. Rejection of this type and/or amount of noise, using a filter, may be challenging and, as such, may result in diminished accuracy in sensing.

(11) Embodiments of the subject matter disclosed herein are directed to a case-driven solution in which a positive voltage is driven onto the case. The inventors have found that driving a voltage onto the case may resolve the electrochemical issues, while reducing the common mode noise that is to be rejected by the sense filter. According to embodiments, the driven voltage may be selected, for example, to be sufficient to stop electrochemical processes such as AEC and to have enough bypass capacitance to keep ripple due to capacitance between the battery and the case at or below a specified level. According to embodiments, the voltage may be driven onto the case throughout the device's operating life, during specific time periods only, and/or the like.

(12) FIG. 1 is a schematic illustration of a system **100** including an IMD **102** implanted within a patient's body **104** and configured to communicate with a receiving device **106**. In embodiments, the IMD **102** may be configured to be implanted subcutaneously within an implantation location or pocket in the patient's chest or abdomen and may be configured to monitor (e.g., sense and/or record) physiological parameters associated with the patient's heart **108**. The IMD **102** may be an implantable cardiac monitor (ICM) (e.g., an implantable diagnostic monitor (IDM), an implantable loop recorder (ILR), etc.) configured to record physiological parameters such as, for example, electrical signals (e.g., electrocardiograms), one or more cardiac activation signals, heart sounds, blood pressure measurements, oxygen saturations, and/or the like. The IMD **102** may be configured to monitor physiological parameters that may include one or more signals indicative of a patient's physical activity level and/or metabolic level, such as an acceleration signal. In embodiments, the IMD **102** may be configured to monitor physiological parameters associated with one or more other organs, systems, and/or the like. The IMD **102** may be configured to sense and/or record at regular

intervals, continuously, and/or in response to a detected event. Such a detected event may be detected by one or more sensors of the IMD **102**, another IMD (not shown), an external device (e.g., the receiving device **106**), and/or the like. In addition, the IMD **102** may be configured to detect a variety of physiological signals that may be used in connection with various diagnostic, therapeutic, and/or monitoring implementations. For example, in embodiments, the IMD **102** may be configured to facilitate cardiac rhythm diagnostics, e.g., by monitoring heart rate, arrhythmias, and/or the like.

(13) For example, the IMD **102** may include sensors or circuitry for detecting respiratory system signals, cardiac system signals, and/or signals related to patient activity. In embodiments, the IMD **102** may be configured to sense intrathoracic impedance, from which various respiratory parameters may be derived, including, for example, respiratory tidal volume and minute ventilation. Sensors and associated circuitry may be incorporated in connection with the IMD **102** for detecting one or more body movement or body posture and/or position related signals. For example, accelerometers and/or GPS devices may be employed to detect patient activity, patient location, body orientation, and/or torso position.

(14) For purposes of illustration, and not of limitation, various embodiments of devices that may be used to record physiological parameters in accordance with the present disclosure are described herein in the context of IMDs that may be implanted under the skin in the chest region of a patient. In embodiments, however, the IMD **102** may include any type of IMD, any number of different components of an implantable system, and/or the like having a housing and being configured to be implanted in a patient's body **104**. For example, the IMD **102** may include a control device, a monitoring device, a pacemaker, an implantable cardioverter defibrillator (ICD), a cardiac resynchronization therapy (CRT) device and/or the like, and may be an implantable medical device known in the art or later developed, for providing therapy and/or diagnostic data about the patient's body and/or the IMD **102**. In various embodiments, the IMD **102** may include both defibrillation and pacing/CRT capabilities (e.g., a CRT-D device).

(15) As shown, the IMD **102** may include a housing **110** having two electrodes **112** and **114** coupled thereto. According to embodiments, the IMD **102** may include any number of electrodes (and/or other types of sensors such as, e.g., thermometers, barometers, pressure sensors, optical sensors, motion sensors, and/or the like) in any number of various types of configurations, and the housing **110** may include any number of different shapes, sizes, and/or features. In embodiments, the IMD **102** may be configured to sense physiological parameters and record the physiological parameters. For example, the IMD **102** may be configured to activate (e.g., periodically, continuously, upon detection of an event, and/or the like), record a specified amount of data (e.g., physiological parameters) in a memory, and communicate that recorded data to a receiving device **106**. In the case of an IDM, for example, the IMD **102** may activate, record cardiac signals for a certain period of time, deactivate, and activate to communicate the recorded signals to the receiving device **106**.

(16) In various embodiments, the receiving device **106** may be, for example, a programmer, controller, patient monitoring system, and/or the like. Although illustrated in FIG. **1** as an external device, the receiving device **106** may include an implantable device configured to communicate with the IMD **102** that may, for example, be a control device, another monitoring device, a pacemaker, an implantable defibrillator, a cardiac resynchronization therapy (CRT) device, and/or the like, and may be an implantable medical device known in the art or later developed, for providing therapy and/or diagnostic data about the patient and/or the IMD **102**.

(17) The system **100** may be used to implement coordinated patient measuring and/or monitoring, diagnosis, and/or therapy in accordance with embodiments of the disclosure. The system **100** may include, for example, one or more patient-internal medical devices, such as an IMD **102**, and one or more patient-external medical devices, such as receiving device **106**. In embodiments, the receiving device **106** may be configured to perform monitoring, and/or diagnosis and/or therapy functions

external to the patient (i.e., not invasively implanted within the patient's body). The receiving device **106** may be positioned on the patient, near the patient, or in any location external to the patient.

(18) In embodiments, the IMD **102** and the receiving device **106** may communicate through a wireless link. For example, the IMD **102** and the receiving device **106** may be coupled through a short-range radio link, such as Bluetooth, IEEE 802.11, and/or a proprietary wireless protocol. The communications link may facilitate uni-directional and/or bi-directional communication between the IMD **102** and the receiving device **106**. Data and/or control signals may be transmitted between the IMD **102** and the receiving device **106** to coordinate the functions of the IMD **102** and/or the receiving device **106**. In embodiments, patient data may be downloaded from one or more of the IMD **102** and the receiving device **106** periodically or on command. The physician and/or the patient may communicate with the IMD **102** and the receiving device **106**, for example, to acquire patient data or to initiate, terminate, or modify recording and/or therapy.

(19) The illustrative system **100** shown in FIG. **1** is not intended to suggest any limitation as to the scope of use or functionality of embodiments of the subject matter disclosed throughout this disclosure. Neither should the illustrative system **100** be interpreted as having any dependency or requirement related to any single component or combination of components illustrated in FIG. **1**. For example, in embodiments, the illustrative system **100** may include additional components. Additionally, any one or more of the components depicted in FIG. **1** can be, in embodiments, integrated with various ones of the other components depicted therein (and/or components not illustrated). Any number of other components or combinations of components can be integrated with the illustrative system **100** depicted in FIG. **1**, all of which are considered to be within the ambit of this disclosure.

(20) FIG. **2A** is a perspective view of an implantable medical device (IMD) **200**, in accordance with embodiments of the disclosure. The IMD **200** may be, or may be similar to, the IMD **102** depicted in FIG. **1**. As shown, the IMD **200** includes a hermetically sealed case, and may include a header **202** arranged at or near a first end **220** of a core assembly **204**. A battery assembly **206** (which may include one or more batteries) is arranged near a second end **224** of the core assembly **204**. The header **202** includes a header housing **202A** that encloses an interior region **202B**. The header **202** may house various circuitry components within its interior, which may include a portion of the operational circuitry of the IMD **200**. The header housing **202A** may contact a patient's bodily tissue when the IMD **200** is subcutaneously implanted in an implantation location or pocket in the patient's chest or abdomen. The interior region **202B** of the header **202** may house circuit components (e.g., an electrode **208** and an antenna **210**) positioned and supported by a scaffold assembly **212**. As shown, the IMD **200** may include, in addition to the electrode **208**, an electrode **214** electrically and/or physically coupled to the battery assembly **206**. In embodiments, the electrode **214** may be integrated with the battery assembly **206**, a housing of the battery assembly **206**, and/or the like. That is, for example, in embodiments, the electrode **214** may be the battery assembly housing, which may also be referred to as a "case." In order to enable sensing of physiological parameters within the patient, the electrode **208** may be positioned to be flush with an interior surface of the housing **202A** of the header **202**. In other instances, the electrode **208** may be positioned by the scaffold assembly **212** to form a portion of an exterior surface of the housing **202A** of the header **202**.

(21) As shown in FIG. **2B**, the core assembly **204** includes a core circuitry assembly **216** enclosed within a core assembly housing **218**. In embodiments, the core assembly housing **218** may be physically and/or electrically coupled to and/or integrated with the battery housing **206A**, in which case the combination of the core assembly housing **218** and the battery housing **206A** may be referred to as a "case." The core circuitry assembly **216** includes operational circuitry configured to perform one or more various functions described herein and may include processing circuitry, input circuitry, output circuitry, communication circuitry, and/or the like. Circuitry may include any

number of different types of electrical and/or logical connections, components, and/or the like, and may include, for example, conductive elements (e.g., wires, conductive traces, etc.), processors (e.g., microprocessors, virtual processors, etc.), and/or other electrical and/or logical components. Thus, circuitry may include hardware, firmware, and/or software. The core assembly housing **218** is coupled, at the first end **220**, to a first feed-through assembly **222**, and coupled, at the second end **224**, to a second feed-through assembly **226**. The feed-through assembly **222** may be configured to provide a throughput for connections configured to connect the circuitry components of the header **202** (e.g., the electrode **208** and the antenna **210**) to the core circuitry assembly **216**. Similarly, the feed-through assembly **226** may be configured to provide a throughput for connections configured to connect one or more batteries (e.g., which are a part of the battery assembly **206**) and/or the electrode **214** to the core circuitry assembly **216**. In embodiments, the combination of the header housing, core assembly housing, battery housing, and feedthrough assembly surface form the case **201**.

(22) As illustrated in FIG. 2A, the core assembly housing **204** includes a first portion **228** configured to be coupled to a second portion **230** along a weld seam **232**. The first portion **228** and second portion **230** may be coupled together by laser welding, seam welding, and/or the like. As shown, for example, in FIGS. 2B and 2C, the first portion **228** of the core assembly housing **218** includes a side wall **234**, a lower wall **236**, and an upper wall **238**. The lower wall **236** and the upper wall **238** each extend, perpendicularly (or at least approximately perpendicularly) in a direction away from an inside surface **234A** of the side wall **234**. As shown, the lower wall **236** is coupled to the side wall **234** by a curved corner portion **240**, and the upper wall **238** is coupled to the side wall **234** by a curved corner portion **242**. In embodiments, the curved corner portions **240** and **242** may be integrated with the lower and upper walls **236** and **238**, respectively, the side wall **234**, and/or the like. That is, for example, the first portion **228** may be a single piece of metal, formed in a press or a mold. In embodiments, the curved corner portions **240** and **242** may be separate components. The curved corner portions **240** and **242** each may be designed to have any desirable radius of curvature. For example, the curved corner portions **240** and **242** each may be configured to have a radius of curvature that provides a desired amount of volume enclosed within the core assembly housing **218**.

(23) As illustrated, for example, in FIGS. 2B and 2C, the lower wall **236** includes a flange **244** that is recessed with respect to an inside surface **246** of the lower wall **236**, and that extends from a first end **248** of the first portion **228** to a second end **250** thereof. The flange **244** may be a thinned portion of the lower wall **236**. In embodiments, the flange **244** may be welded to the lower wall **236**. Similarly, the upper wall **238** includes a flange **252** that is recessed with respect to an inside surface **254** of the upper wall **238**, and that extends from the first end **248** of the first portion **228** to the second end **250** thereof. The flange **252** may be a thinned portion of the upper wall **238**. In embodiments, the flange **252** may be welded to the upper wall **238**.

(24) As is also shown, for example, in FIGS. 2B and 2C, the second portion **230** of the core assembly housing **218** includes a side wall **256**, a lower wall **258**, and an upper wall **260**. The lower wall **258** and the upper wall **260** each extend, perpendicularly (or at least approximately perpendicularly) in a direction away from an inside surface **256A** of the side wall **256**. As shown, the lower wall **258** is coupled to the side wall **256** by a curved corner portion **262**, and the upper wall **260** is coupled to the side wall **256** by a curved corner portion **264**. In embodiments, the curved corner portions **262** and **264** may be integrated with the lower and upper walls **258** and **260**, respectively, the side wall **256**, and/or the like. That is, for example, the second portion **230** may be a single piece of metal, formed in a press or a mold. In embodiments, the curved corner portions **262** and **264** may be separate components. The curved corner portions **262** and **264** each may be designed to have any desirable radius of curvature such as, for example, a radius of curvature that is identical or similar to the radius of curvature of each of the curved corner portions **240** and **242**. For example, the curved corner portions **262** and **264** each may be configured to have a radius of

curvature that provides a desired amount of volume enclosed within the core assembly housing **218**.

(25) As illustrated, for example, in FIGS. **2B** and **2C**, the lower wall **258** includes a flange **266** that is recessed with respect to an outside surface **268** of the lower wall **258**, and that extends from a first end **270** of the second portion **230** to a second end **272** thereof. The flange **266** may be a thinned portion of the lower wall **258**. In embodiments, the flange **266** may be welded to the lower wall **258**. Similarly, the upper wall **260** includes a flange **274** that is recessed with respect to an outside surface **276** of the upper wall **260**, and that extends from the first end **270** of the second portion **230** to the second end **272** thereof. The flange **274** may be a thinned portion of the upper wall **260**. In embodiments, the flange **274** may be welded to the upper wall **260**.

(26) The core assembly housing **218** may also include notches **278** defined in the first and second ends **248** and **250**, respectively, of the first portion **228**, and extending from the inside surface **234A** to the outside surface **234B** of the side wall **234**. Similarly, the core assembly housing **218** may also include notches **280** defined in the first and second ends **270** and **272**, respectively, of the second portion **230**, and extending from the inside surface **256A** to the outside surface **256B** of the side wall **256**. The notches **278** and **280** may be an artifact of a progressive die manufacturing process in which the first and second portions **228** and **230** of the core assembly housing **218** are produced in a continuous strip and formed into shape in successive operations. The notches **278** and **280** may be left when the first and second portions **228** and **230** are broken away from the strip. In embodiments, the strip may be configured such that the notches are small enough to be consumed in the weld pool when the core assembly housing **218** is welded to the first and second feedthrough assemblies **222** and **226**. For example, in embodiments, the notches **278** and **280** may extend into the portions **228** and **230** by less than or equal to approximately 0.003 inches.

(27) The illustrative IMD **200** shown in FIGS. **2A-2C** is not intended to suggest any limitation as to the scope of use or functionality of embodiments of the subject matter disclosed throughout this disclosure. Neither should the illustrative IMD **200** be interpreted as having any dependency or requirement related to any single component, feature, or combination of components or features illustrated in FIGS. **2A-2C**. For example, in embodiments, the illustrative IMD **200** may include different and/or additional components and/or features. Any number of other components, features, or combinations of components or features can be integrated with the illustrative IMD **200** depicted in FIGS. **2A-2C**, all of which are considered to be within the ambit of this disclosure. Additionally, any one or more of the components and/or features depicted in FIGS. **2A-2C** can be, in embodiments, integrated with various ones of the other components and/or features depicted therein (and/or components and/or features not illustrated).

(28) Moreover, as used herein, the terms “side wall,” “lower wall,” “upper wall,” “upward,” and “downward” are used to refer to the specific features to which they refer, but are characterized in the context of the illustrations for clarity and to describe relative orientations of features with respect to other features, and are not intended to imply any particular orientation of the IMD **200**, or absolute (or preferred) orientations of features thereof. That is, for example, even if the IMD **200** were to be rotated around a longitudinal axis such that the outer surface **234B** of the side wall **234** was parallel to a horizontal plane, the side wall **234** would still be referred to, for the purposes of this disclosure, as a “side wall.”

(29) FIG. **3** is a functional schematic diagram of circuitry **300** associated with an IMD, according to a case-negative design, in accordance with the prior art. The circuitry **300** depicted in FIG. **3** may be associated with the sensing of physiological signals by two electrodes **302** and **304**. The circuitry **300** includes a battery **306** that is enclosed within a case **308**, to which the electrode **304** is electrically coupled. In embodiments, the case **308** (or a portion thereof) may function as the electrode **304**. Supply rails **310** and **312** define the current path for delivery of energy to operational circuitry **314**, and are electrically coupled to the battery **306**.

(30) In operation, the electrodes **302** and **304** obtain a physiological signal from a patient, and

transmit that obtained physiological signal to an input component **316** (e.g., a sense amplifier). As shown in FIG. **3**, the electrode **304** is directly coupled to a first input node of the input component **316** via a conductive element **318** (e.g., one or more conductive traces and/or wires), while the electrode **302** is directly coupled, via a conductive element **320** (e.g., one or more conductive traces and/or wires), to a second input node of the input component **316**.

(31) According to embodiments, the conductive element **318** is directly coupled to the negative (or, in embodiments, positive) terminal **322** (anode) of the battery **306**, and to the common (ground) reference (not shown) of the operational circuitry **314**. In embodiments, the ground reference may be used as a Kelvin Connection, joining the battery to the sensing electrode **304**. As shown, a capacitance (e.g., caused by the battery electrolyte coupling to the battery case) **324** may be disposed between the positive terminal **326** and the case **308**. As is further depicted in FIG. **3**, the output node of the input component **316** is coupled to processing circuitry **328**. The processing circuitry **328** may include any number of different types of processing components, including, for example, one or more microprocessors, circuit components, and/or the like. In the case-negative design depicted in FIG. **3**, the case **308** is shorted to the negative battery terminal **322**. In embodiments, doing so removes battery noise from sensed signals, but creates electrochemical issues that may cause damage to the case **308**, the battery **306**, and/or other components of the IMD **300**.

(32) FIG. **4** is a functional schematic diagram of circuitry **400** associated with an IMD, employing a case-positive design, in accordance with the subject matter disclosed herein. The circuitry **400** depicted in FIG. **4** may be associated with the sensing of physiological signals by two electrodes **402** and **404**. The circuitry **400** includes a battery **406** that is enclosed within a case **408**, to which the electrode **404** is electrically coupled. In embodiments, the case **408** (or a portion thereof) may function as the electrode **404**. Supply rails **410** and **412** define the current path for delivery of energy to operational circuitry **414** and are electrically coupled to the battery **406**.

(33) Supply rails **410** and **412** define the current path for delivery of energy to operational circuitry **414**. The positive supply rail **412** is electrically coupled to the battery **406** through the case **408** (or a portion thereof), and the negative supply rail **410** is electrically coupled directly to the battery **406**. As shown, a capacitance **416** may be disposed between the negative terminal **418** and the case **408**. In operation, the electrodes **402** and **404** obtain a physiological signal from a patient, and transmit that obtained physiological signal to an input component **420** (e.g., a sense amplifier). As shown in FIG. **4**, the electrode **404** is directly coupled to a first input node of the input component **420** via the case **408** and a conductive element **422**, while the electrode **402** is directly coupled, via a conductive element **424** (e.g., one or more conductive traces and/or wires), to a second input node of the input component **420**. As is further depicted in FIG. **4**, the output node of the input component **420** is coupled to processing circuitry **426**.

(34) As can be seen from inspection of FIGS. **3** and **4**, the circuit configuration depicted in FIG. **4** may be similar to the circuit configuration depicted in FIG. **3**, with the exception of the configuration of the components that short a battery terminal to the case. In the case-negative design of FIG. **3**, the case is shorted to the negative battery terminal, while, in the case-positive design of FIG. **4**, the case is shorted to the positive battery terminal. In the case-positive design, the electrochemical issues of the case-negative design are resolved, but a battery voltage ripple is injected as a common mode noise. Rejecting this noise with the sense filter can be challenging. For example, voltage ripple from the battery may be superimposed on a sense signal sensed between the can and the header electrode, and may be tens of thousands of times the magnitude of the sense signal, which may be only several hundred microvolts.

(35) The illustrative circuitry **400** shown in FIG. **4** is not intended to suggest any limitation as to the scope of use or functionality of embodiments of the subject matter disclosed throughout this disclosure. Neither should the illustrative circuitry **400** be interpreted as having any dependency or requirement related to any single component, feature, or combination of components or features

illustrated in FIG. 4. For example, in embodiments, the illustrative circuitry **400** may include different and/or additional components and/or features. Any number of other components, features, or combinations of components or features can be integrated with the illustrative circuitry **400** depicted in FIG. 4, all of which are considered to be within the ambit of this disclosure. Additionally, any one or more of the components and/or features depicted in FIG. 4 can be, in embodiments, integrated with various ones of the other components and/or features depicted therein (and/or components and/or features not illustrated).

(36) FIG. 5 is a functional schematic diagram of circuitry **500** associated with an IMD, employing a case-driven design, in accordance with the subject matter disclosed herein. The IMD may be, include, or be included in, the IMD **102** depicted in FIG. 1 and/or the IMD **200** depicted in FIGS. 2A-2C. According to embodiments, in the case-driven design, a regulated voltage is driven to the case. This design may facilitate mitigating the electrochemical issues of the case-negative design, while avoiding the challenging common mode noise of the case-positive design.

(37) The circuitry **500** depicted in FIG. 5 may be associated with the sensing of physiological signals by two electrodes **502** and **504**. The circuitry **500** includes a battery **506** that is enclosed within a case **508**, to which the electrode **504** is electrically coupled. In embodiments, the case **508** (or a portion thereof) may function as the electrode **504**. Supply rails **510** and **512** define the current path for delivery of energy to operational circuitry **514** and are electrically coupled to the battery **506**.

(38) Supply rails **510** and **512** define the current path for delivery of energy to operational circuitry **514**. As shown, the negative battery terminal **516** and the positive battery terminal **518** are isolated from the case **508** by a respective capacitance **520**, **522**. The positive supply rail **512** is electrically coupled directly to the battery **506** and the negative supply rail **510** is electrically coupled directly to the battery **506**. In operation, the electrodes **502** and **504** obtain a physiological signal from a patient, and transmit that obtained physiological signal to an input component **524** (e.g., a sense amplifier). As shown in FIG. 5, the electrode **504** is directly coupled to a first input node of the input component **524** via the case **508** and a conductive element **526**, while the electrode **502** is directly coupled, via a conductive element **528** (e.g., one or more conductive traces and/or wires), to a second input node of the input component **524**. As is further depicted in FIG. 4, the output node of the input component **524** is coupled to processing circuitry **530**.

(39) As shown in FIG. 5, the operating circuitry **514** includes a number of different components such as, for example, a radio frequency (RF) component **532**, a digital processing component **536** (DIG), a DIG voltage regulator (VDIG) **534**, and a driving voltage regulator (VREG) **538**. In embodiments, the functions of the RF component **532**, the DIG **534**, and the VDIG **536** are not germane to the present disclosure, but are illustrated as examples of various types of operational components might be present in some implementations of embodiments of the subject matter disclosed herein.

(40) According to embodiments, the driving voltage regulator (VREG) **538** may be configured to drive a voltage onto the can **508**. In embodiments, VREG **538** may be a voltage regulator circuit component, a number of voltage regulator circuit components, a processor, logic, and/or the like. VREG **538** may, as illustrated in FIG. 5, also be configured to drive the power supply for the input component **524** (e.g., sense amp). Using a common power source to drive the case and to power the sense circuit may also improve the sense circuit common mode rejection ratio. Even though the case is being driven there will still be some voltage ripple on the case from the battery. This voltage ripple will be a common mode noise signal that the sense circuit needs to reject. When the same voltage source is used to power the sense filters and drive the case some factors affecting common mode rejection such as transistor body capacitance can be canceled out. According to embodiments, the circuit power path can be common (coupled) with the circuit input from the case if the common net impedance is kept small enough to prevent voltage noise (due to current×impedance) from exceeding the desired noise floor for the system.

(41) In other embodiments, a separate voltage regulator may be provided for driving the sense amp. In the embodiments illustrated in FIG. 5, the driven voltage is generated by regulating down a voltage from the battery 506, generating a quieter voltage that is configured to mitigate or stop the electrochemical issues, while having enough bypass capacitance or drive strength to mitigate ripple due to capacitance between the battery 506 and the case 508.

(42) In embodiments, for example, the driven voltage may be configured to be approximately the same as the power supply for the sense circuit. That is, for example, the driven voltage may be selected to be approximately equal to the positive power supply. In embodiments, for example, the voltage on the positive side of the circuitry 500 may be approximately 3 V, and the driven voltage may be between approximately 1 and 2.5 V (e.g., 1 V, 2.2 V, etc.). In embodiments, the driven voltage may be selected to be large enough for producing a signal that is sufficient for mitigating the electrochemical reaction while producing a reliable and accurate sense signal. In embodiments, for example, the VREG 326 may be configured to be within 7/10 of a volt of the positive power supply. According to embodiments, VREG 538 may include any number of circuit components and may be configured to generate a power supply for the can 508 that, from an AC loading perspective, follows the negative terminal battery voltage. In embodiments, the driven voltage may be higher than the positive battery terminal using an amplifying circuit such as a charge pump. This may be desirable for biasing certain circuit inputs at a higher voltage. In contrast, for example, to the case-positive design depicted in FIG. 4, in which the sense amp does all of the work in rejecting the noise, in the case-driven design depicted in FIG. 5, VREG 538 does most of the work.

(43) According to embodiments, the case-driven design may be implemented for driving a voltage on to a case according to any number of different schedules, arrangements, and/or the like. For example, in embodiments, the driven voltage may be held at a constant voltage such as, e.g., 2.2 V. In embodiments, the driven voltage can be controlled by device state. For example, while in shelf/storage mode, the device may be configured to either float case or drive to positive voltage such as shorting to the positive battery terminal. When entering an active mode such as sensing, the IMD can be configured to drive the case to a regulated voltage. According to other embodiments, the driven voltage can be controlled to match the battery discharge state. For example, the case may be driven to 2.5 V early in the cell state but then gradually reduced to 2.2 V as the battery is discharged and the average unloaded battery voltage decreases. This may have the advantage of keeping the driven voltage high to maintain a small voltage difference between the case and the battery anode while protecting against regulation issues as the battery voltage decreases.

(44) In embodiments, the drive circuit may change operating mode based on device activity such as using a high-power mode when significant battery ripple is expected due to device activity. This may facilitate minimizing the energy consumed by the drive circuit over the life of the device. In embodiments, the case drive may be suspended during device activities that require driving the case to a different voltage or sensing case currents or voltages. For example, the case drive may be disconnected (tri-stated) when device therapy operation such as a pace pulse or impedance measurement current pulse is being driven through the case. In other words, embodiments include releasing the drive to the can for short periods of time when not sensing.

(45) For example, in embodiments, the device may be configured to drive a voltage onto the case only during a certain period of the life of the device. In embodiments, noise is much more significant near a battery's end of life, as internal resistance of the battery changes as a function of depth of discharge. Thus, near the end of the battery's life, the case may be driven with a regulated voltage. In contrast, for example, in a storage mode before the device is implanted and active, the sensing circuit is off, and, thus, the presence of ripple is not an issue. Thus, in embodiments, while in the storage mode, the device may be shorted to the positive (as in a case-positive design), or let it float to the positive, which would mitigate the electrochemical issues. Once the device enters and active state in which it is sensing, the can may be configured in a case-driven design or a case-negative design.

(46) For example, in embodiments, rather than implementing a case-driven design, the IMD may be configured to be shorted to positive or floated to positive during the storage mode, and placed in a case-negative design upon activation and/or implantation. This mode of operation may facilitate reducing the time and temperature exposure to limit the total electrochemical reaction that occurs. It also may facilitate reducing the amount of electrochemical reaction that occurs (by limiting both time and temperature exposure to a fully charged battery) while still having a simple drive circuit and reducing case ripple voltage while sensing.

(47) The illustrative circuitry **500** shown in FIG. 5 is not intended to suggest any limitation as to the scope of use or functionality of embodiments of the subject matter disclosed throughout this disclosure. Neither should the illustrative circuitry **500** be interpreted as having any dependency or requirement related to any single component, feature, or combination of components or features illustrated in FIG. 5. For example, in embodiments, the illustrative circuitry **500** may include different and/or additional components and/or features. Any number of other components, features, or combinations of components or features can be integrated with the illustrative circuitry **500** depicted in FIG. 5, all of which are considered to be within the ambit of this disclosure.

Additionally, any one or more of the components and/or features depicted in FIG. 5 can be, in embodiments, integrated with various ones of the other components and/or features depicted therein (and/or components and/or features not illustrated).

(48) Various modifications and additions can be made to the exemplary embodiments discussed without departing from the scope of the disclosed subject matter. For example, while the embodiments described above refer to particular features, the scope of this disclosure also includes embodiments having different combinations of features and embodiments that do not include all of the described features. That is, for example, embodiments may include one or more filters and/or other components that facilitate interpreting sensed physiological signals in the presence of some interference caused by having an energy supply current sharing a portion of the physical sense pathway. Accordingly, the scope of the disclosed subject matter is intended to embrace all such alternatives, modifications, and variations as fall within the scope of the claims, together with all equivalents thereof.

Claims

1. A medical device comprising: a case, wherein at least a portion of the case functions as a first sensing electrode; a second sensing electrode; a battery positioned within the case and including a positive battery terminal and a negative battery terminal; and a first capacitor electrically coupled between the case and the negative battery terminal.
2. The medical device of claim 1, wherein the case is shorted to the positive battery terminal.
3. The medical device of claim 1, further comprising a second capacitor electrically coupled between the case and the positive battery terminal.
4. The medical device of claim 1, wherein the first sensing electrode and the second sensing electrode are electrically coupled to an amplifier.
5. The medical device of claim 4, wherein a sense signal sensed between the first sensing electrode and the second sensing electrode is transmitted to the amplifier.
6. The medical device of claim 5, wherein the sense signal is no greater than several hundred microvolts.
7. The medical device of claim 4, wherein the battery is electrically coupled to the amplifier.
8. The medical device of claim 4, wherein the amplifier is electrically coupled between processing circuitry and either or both of the first sensing electrode and the second sensing signal.
9. The medical device of claim 1, further comprising an integrated circuit programmed to cause a voltage to be driven to the case.
10. The medical device of claim 9, wherein the integrated circuit is programmed to drive the

voltage at a level that is selected based on a battery discharge state.

11. The medical device of claim 9, wherein the voltage is 1-2.5 volts.

12. The medical device of claim 9, wherein the voltage is within 7/10 volts of a battery voltage of the battery.

13. The medical device of claim 9, wherein the voltage is approximately equal to a battery voltage of the battery.

14. The medical device of claim 9, wherein the voltage is held constant.

15. The medical device of claim 9, wherein the voltage is at a level to create a voltage difference between the case and the positive battery terminal.

16. The medical device of claim 9, further comprising an amplifying circuit for increasing a level of the voltage.

17. The medical device of claim 1, wherein the second sensing electrode is coupled to a header, which is coupled to the case.

18. The medical device of claim 1, further comprising means for reducing common mode noise in a sensing signal generated by the first sensing electrode and the second sensing electrode.

19. A medical device comprising: a case, wherein at least a portion of the case functions as a first sensing electrode; a second sensing electrode; a battery positioned within the case and including a positive battery terminal and a negative battery terminal; and means for limiting chemical reactions with the case.

20. The medical device of claim 19, further comprising means for reducing common mode noise in a sensing signal generated by the first sensing electrode and the second sensing electrode.
